LAB EXPERIMENT RECORD
13-10-2026---Set 17: Employee Performance Analysis

Department: AIML
Course Code / Course Name: DSA0601 / Data Handling and Visualization
Experiment: Data Visualization using R and Tableau
Name: Dudekula Mohammad Ilyas
Register Number: 192325017
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Aim

To analyze employee performance data using a line chart, bar chart and table, and to build an interactive Tableau dashboard for exploring employee performance.

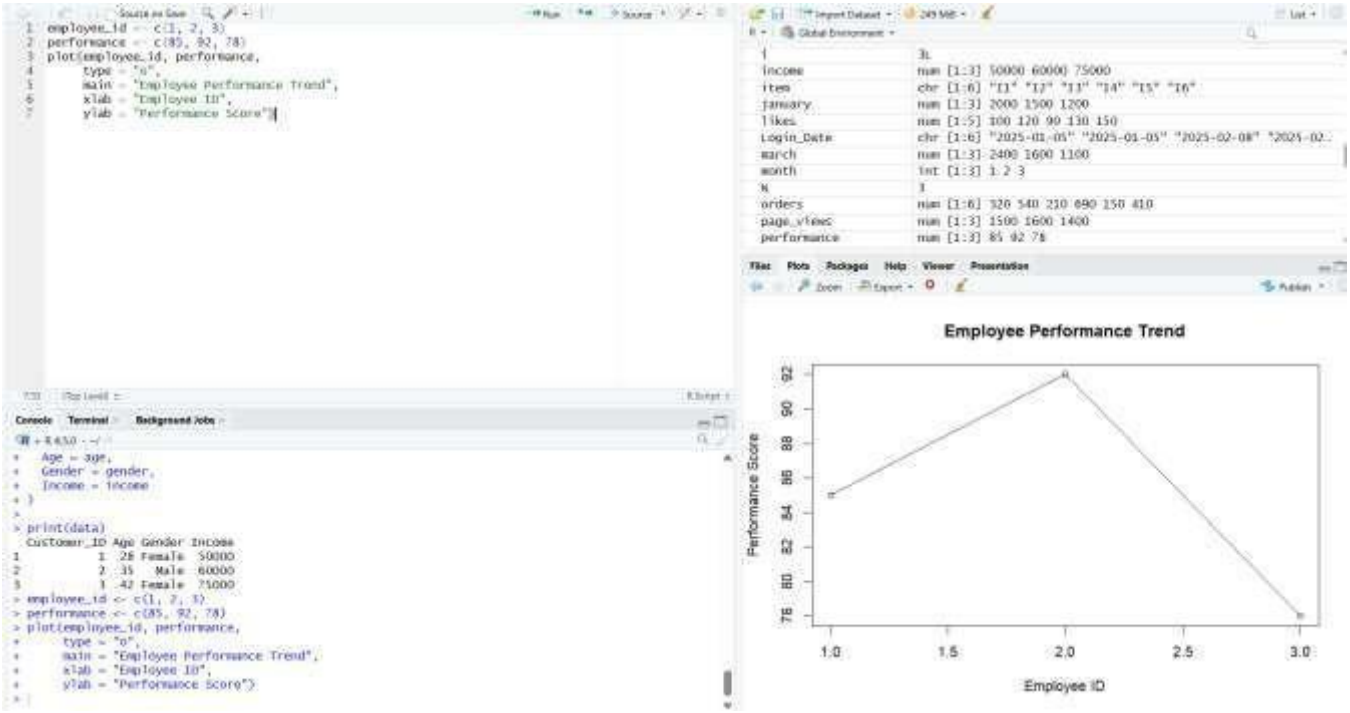
Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

Question 1: Line Chart of Performance Trend

Question: Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

Aim: To visualize the performance trend of employees using a line chart with labelled axes and a title.

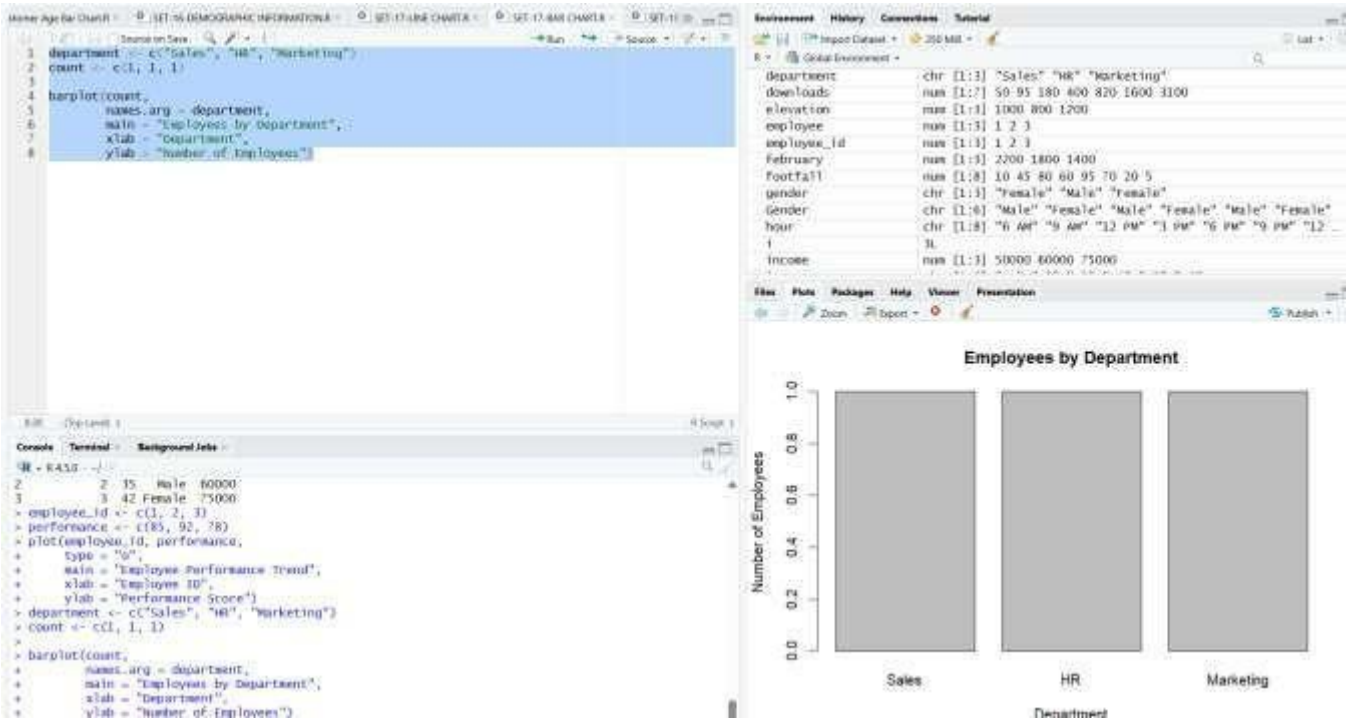


Result: Performance rises from 85 for Employee 1 to a peak of 92 for Employee 2, then falls to the lowest value of 78 for Employee 3.

Question 2: Bar Chart of Employees by Department

Question: Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

Aim: To show the distribution of employees across departments using a labelled bar chart.

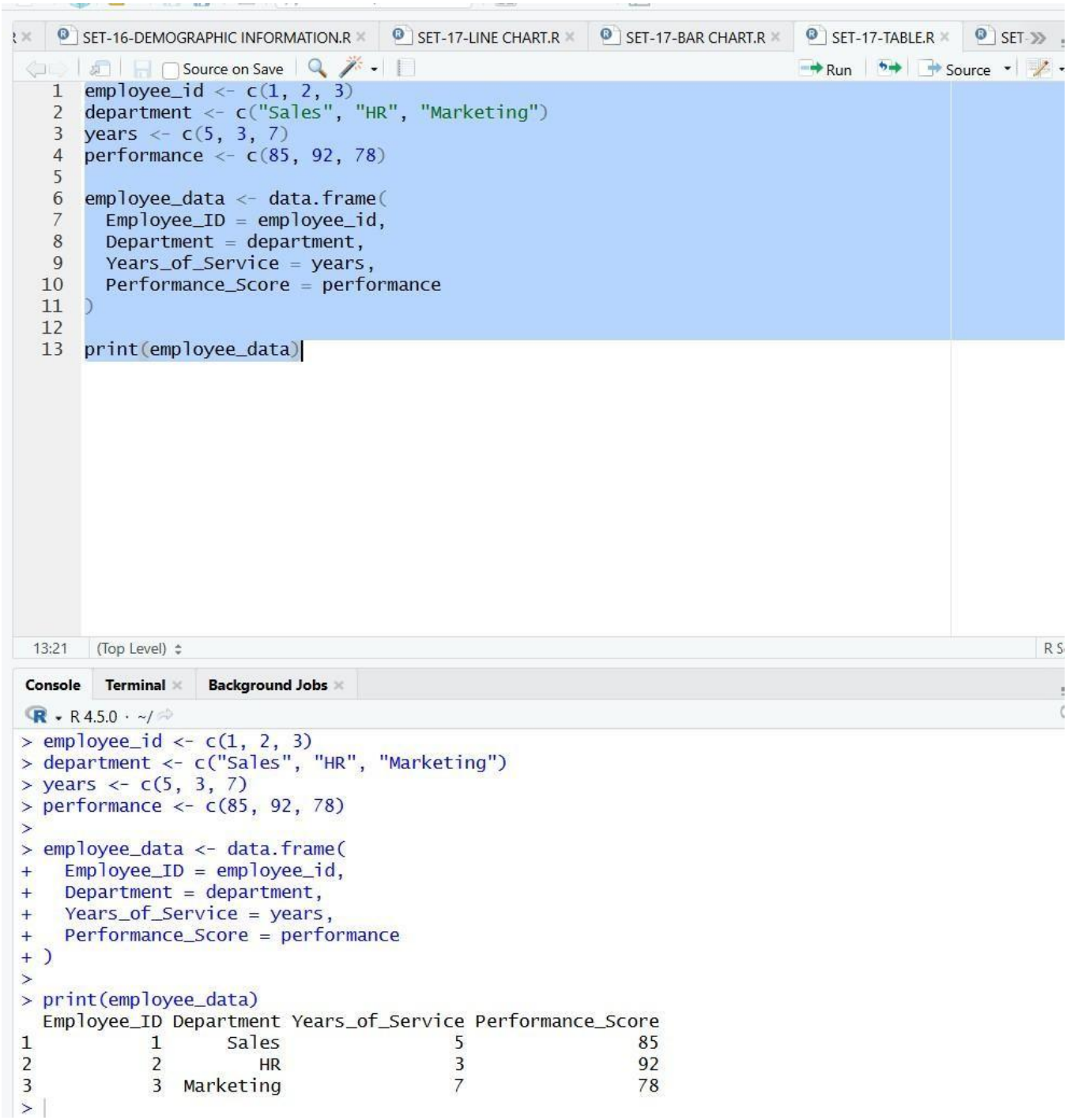


Result: Sales, HR and Marketing each have exactly 1 employee, so the three bars are equal in height.

Question 3: Table of Employee Performance

Question: Build a table to display the performance data for each employee. Label the table elements.

Aim: To present the department, years of service and performance score of each employee in a labelled table.



Result: The table lists all three employees; the HR employee has the highest score (92), and the Marketing employee, with the most service (7 years), has the lowest (78).

Question 4: Interactive Tableau Dashboard

Question: Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.

Aim: To build an interactive Tableau dashboard that combines the line chart, bar chart and table for exploring employee performance.



Employee Details

Employee ID	Name	Department	Years of Service	Performance Score
E001	John	HR	2	75
E002	Emma	Finance	4	82
E003	Liam	IT	1	68
E004	Olivia	Marketing	6	90
E005	Noah	Operations	3	78
E006	Ava	IT	5	85
E007	Lucas	Finance	2	72
E008	Mia	HR	4	88

Result: The dashboard places all three views together, and a department filter updates them so users can explore performance interactively.

Overall Result

Observation: The average performance score is 85, and a longer service period does not guarantee a higher score in this small sample.